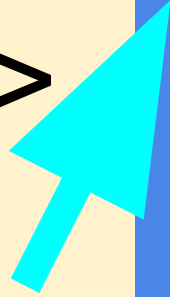


ONE PERSON IN GROUP-> CLICK USE TEMPLATE



WHEN THE COPIED TEMPLATE OF THESE SLIDES OPENS IN A NEW TAB:

- ADD YOUR Group NAME TO THE TITLE
- SHARE WITH GROUP MEMBERS (USE SHARE BUTTON)
- COPY THE TITLE AND LINK TO YOUR RECORDS (Digital folder/table of contents)
- Remember it is advisable to make a PERSONAL copy and link it when DONE

Introduction to Solutions

ESSENTIAL QUESTIONS

Remember, you should be able to thoroughly answer these AFTER learning through following resources.

- What properties could solutes have that allows them to dissolve in solvents?
- What the different classes of solutions?
- What role does intermolecular forces play in solubility of a solute?

Instructions

Follow through the **Engage to Apply** resources in order .

- Once complete, choose from Extend resources as interested.
- Record your learning in **these slides- Remember to keep track of where to find these slides for studying!**
- Complete the **Essential Questions Response in the Apply Section** .
- Try the Extend section and optional Apply practice as desired to deepen your understanding

Hyperdoc Elements

Click for quick access to:

- [Engage](#)
- [Explore](#)
- [Explain](#)
- [Apply](#)
- [Extend](#)

Terminology - As you encounter terms new to you, define in your own words here, include examples (make them up!)



Part 1: ENGAGE

Engage - Part A: 5 minutes

In your groups, pick any 3 below. Discuss what you know, and what questions you have.

- The terms melting and dissolving have the same meaning
- Oil doesn't mix with water because oil and water molecules repel each other
- In a solution, solute is the component present in the smallest quantity
- Solutions only refer to liquids
- Solubility is affected by temperature and pressure
- When diluting acids, water should be ADDED TO the concentrated acid

Engage - Part B: 5 minutes

In your groups, brainstorm 5 reasons solutions are important.

Give as many reasons as you can, whether you think they are correct or not! (don't self edit!)

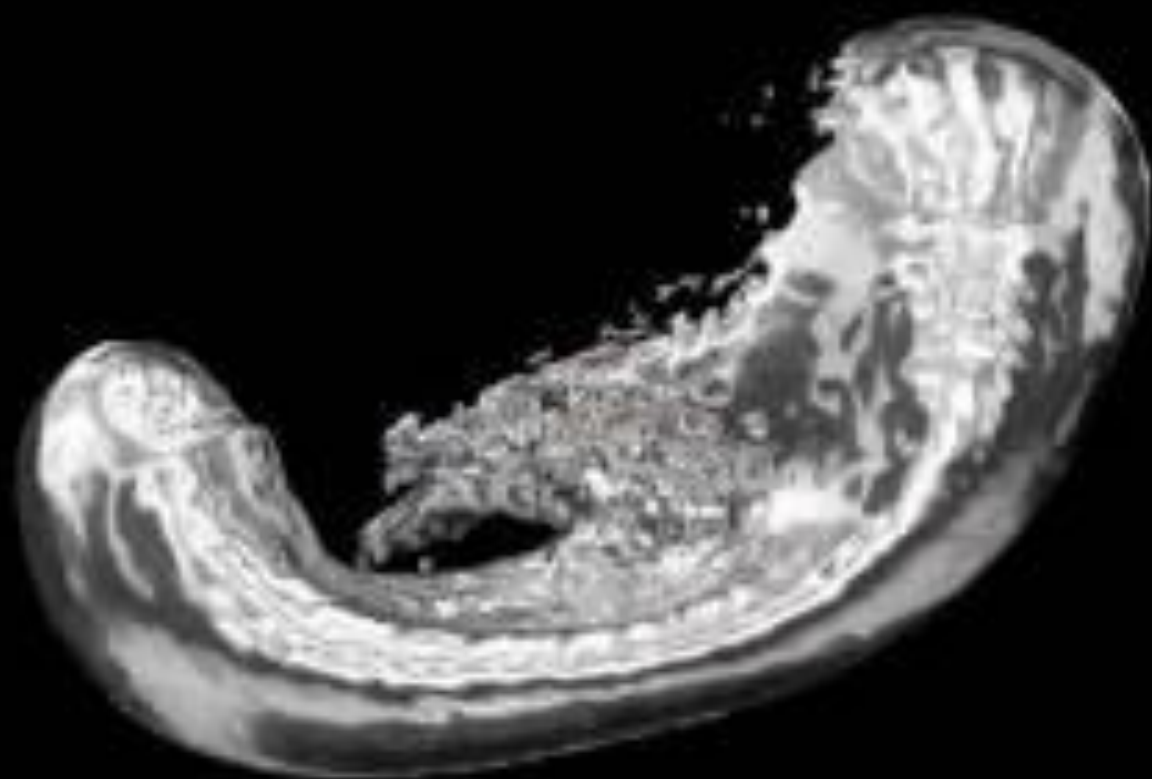
Engage - Part C: 5 minutes

In your groups, watch the videos on slide 11 and slide 12. After the second, briefly imagine what that technology will change in the future (2 mins).

Scene From Terminator: T-1000

Is this Possible?





Part 2: EXPLORE

Explore - Classes of Solutions

A solution is made up of a solute (the material that dissolves) and a solvent, (the material that does the dissolving). There are different ways we can form a solution (classes), and it exists in all three states of matter (solid solutions, liquid solutions, gas solutions)

Table 14.1 Types of Solutions

Solute	Solvent	Solution
<i>Gaseous Solutions</i>		
gas	gas	air (nitrogen, oxygen, argon gases)
liquid	gas	humid air (water vapor in air)
<i>Liquid Solutions</i>		
gas	liquid	carbonated drinks (CO_2 in water)
liquid	liquid	vinegar ($\text{HC}_2\text{H}_3\text{O}_2$ in water)
solid	liquid	salt water (NaCl in water)
<i>Solid Solutions</i>		
liquid	solid	dental amalgam (Hg in Ag)
solid	solid	sterling silver (Cu in Ag)



Explore - Polarity

Recall: Determining Solubility

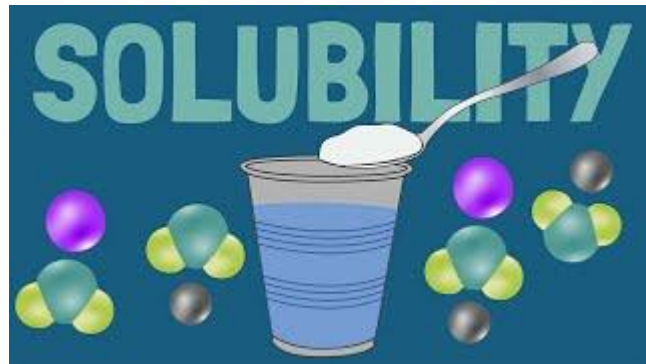


Solubility of a solute in some solvent is dependent on the intermolecular (IM) forces within both compounds:

- The forces that attract Solute-Solute and Solute-Solvent

Generally, if the IM forces in the solvent are the same (or similar) to those within the solute, we can say the solute is soluble in that solvent

- This rule can be summarized as LIKE DISSOLVES LIKE



Explore - Polarity



General Solubility Guidelines

LIKE DISSOLVES LIKE

		Solute	
		Polar	Non-Polar
Solvent	Polar	Soluble	Insoluble
	Non-Polar	Insoluble	Soluble

*It is important to remember that these rules are **general**, and there are many exceptions. We can treat them as **guidelines**.*



Explore - Polar & Non-Polar Solvents

- A liquid composed of polar molecules is a **polar solvent**. Water and ethanol are polar solvents.
- A liquid composed of nonpolar molecules is a **nonpolar solvent**. Hexane is a nonpolar solvent.

Table 14.3 Selected Polar and Nonpolar Solvents

Polar Solvents	Nonpolar Solvents
water, H_2O	hexane, C_6H_{14}
methanol, CH_3OH	heptane, C_7H_{16}
ethanol, $\text{C}_2\text{H}_5\text{OH}$	toluene, C_7H_8
acetone, $\text{C}_3\text{H}_6\text{O}$	carbon tetrachloride, CCl_4
methyl ethyl ketone, $\text{C}_4\text{H}_8\text{O}$	chloroform, CHCl_3
formic acid, HCHO_2	methylene chloride, CH_2Cl_2
acetic acid, $\text{HC}_2\text{H}_3\text{O}_2$	ethyl ether, $\text{C}_4\text{H}_{10}\text{O}^*$

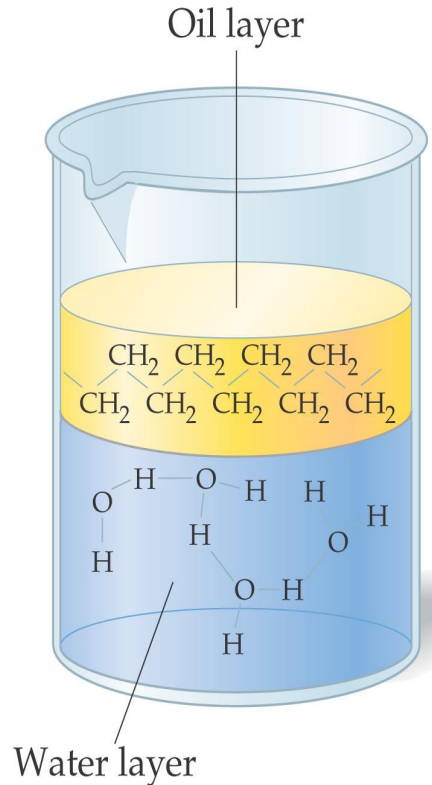
*The general rule that oxygen-containing solvents are polar has some exceptions. For example, ethyl ether contains oxygen and is a nonpolar solvent.

- Methanol dissolves in water but hexane does not dissolve in water.
- Hexane dissolves in toluene, but water does not dissolve in toluene.

Explore - Miscible & Immiscible



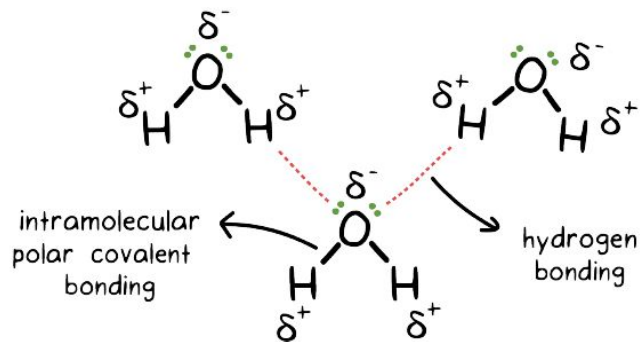
- Two liquids that completely dissolve in each other are *miscible* liquids.
- Two liquids that are not miscible in each other are *immiscible* liquids.
- Polar water and nonpolar oil are immiscible liquids and do not mix to form a solution.



Explore - Solids in Solution



- When a solid substance dissolves in a liquid, the solute particles are attracted to the solvent particles.
- When a solution forms, the solute particles are more strongly attracted to the solvent particles than other solute particles.
- We can also predict whether a solid will dissolve in a liquid by applying the *like dissolves like* rule.

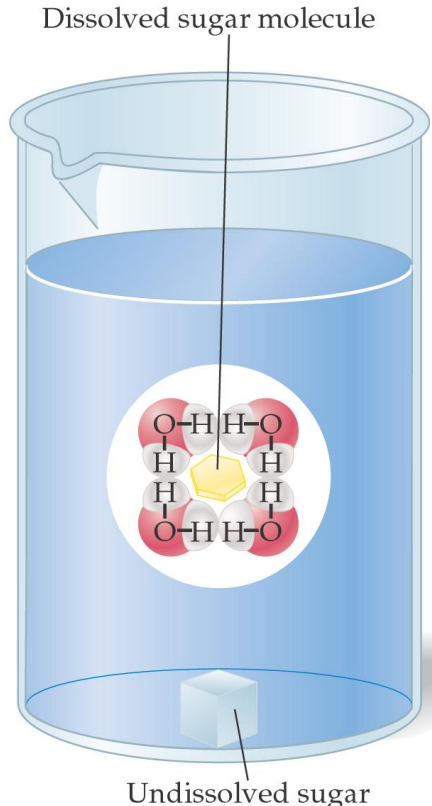




Explore - Solubility of Ionic Compounds

- Ionic compounds, like sodium chloride, are soluble in polar solvents and insoluble in nonpolar solvents.
- Polar compounds, like table sugar ($C_{12}H_{22}O_{11}$), are soluble in polar solvents (dipole and H-bond attractions) and insoluble in nonpolar solvents.
- Nonpolar compounds, like naphthalene ($C_{10}H_8$), are soluble in nonpolar solvents and insoluble in polar solvents. LDF allow non-polars to attract and mix but cannot disrupt stronger dipole attractions

Explore - Dissolving Process

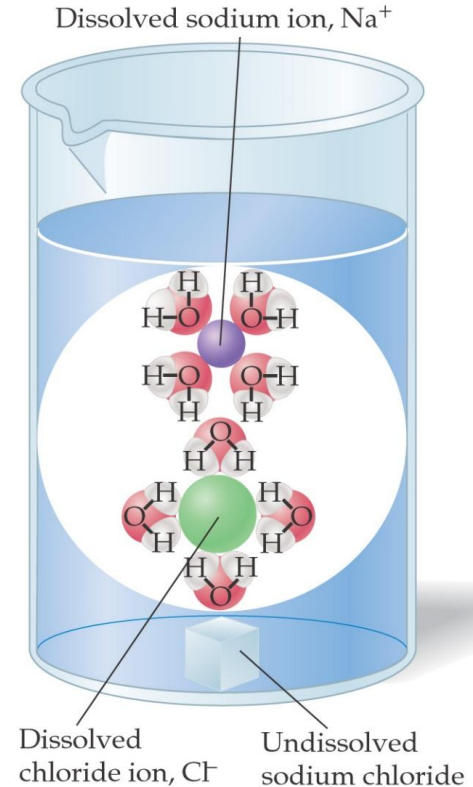


- When a soluble crystal is placed into a solvent, it begins to dissolve.
- When a sugar crystal is placed in water, the water molecules attack the crystal and begin pulling part of it away and into solution.
- The sugar molecules are held within a cluster of water molecules called a *solvent cage*.

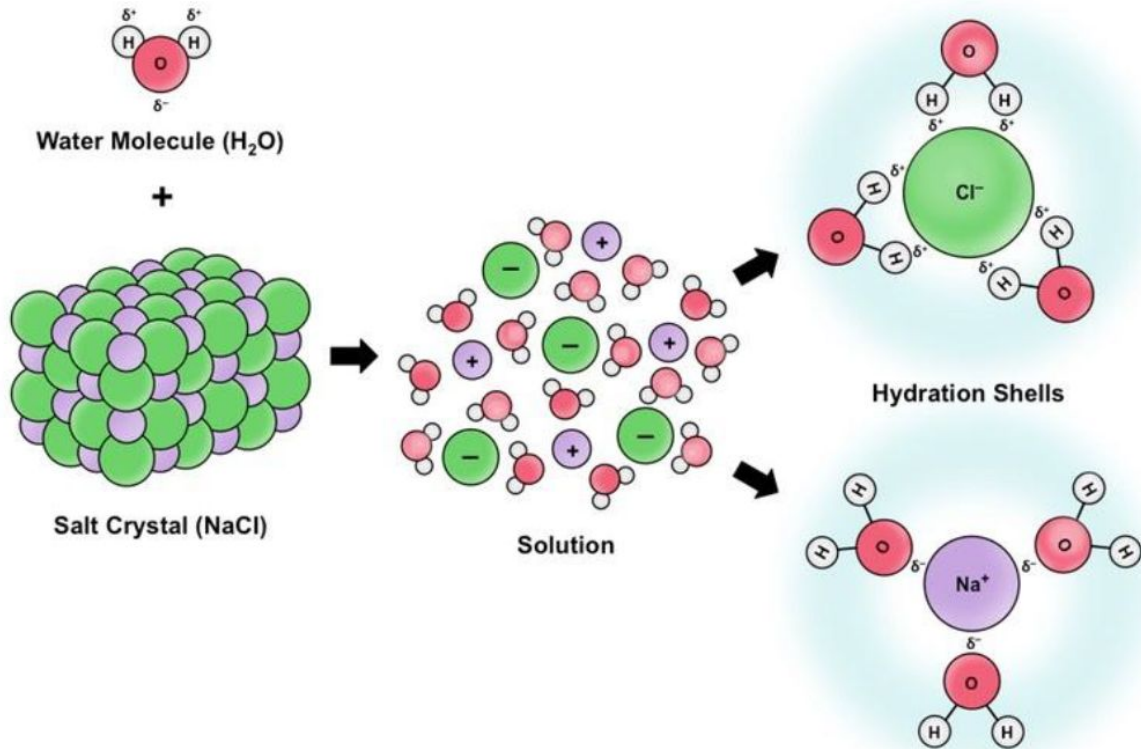
Explore - Dissolving Process



- When a sodium chloride crystal is placed in water, the water molecules attack the edge of the crystal.
- In an ionic compound, the water molecules pull individual ions off of the crystal.
- The anions are surrounded by the positively charged hydrogens on water.
- The cations are surrounded by the negatively charged oxygen on water.



Explore - Dissolving Process



Explore - Dissolving



IN YOUR GROUP - WORK AT A CLASS WHITEBOARD

Record your work (take a photo and insert in these slides, redraw it in your notes etc)

1. Draw a three NaCl formula units, showing the ion charges (3 Na ions with three Cl ions, a small crystal!)
2. Draw a water molecule, showing its bent shape and polarity of the bond (δ^- or δ^+)
3. Draw what will happen when NaCl is put into the water (THINK - use your knowledge of intermolecular forces and attraction)
4. Show your teacher & record!
5. Replace the NaCl with NH_3 and repeat Step 3 & 4. Draw the shape and polarity of NH_3 !

Explore - Dissolving

Slide to insert images of your board work

Explore - Dissolving



{Optional} For further help you can watch this video

The video thumbnail shows a person's hands holding a ball-and-stick model of a crystal lattice. The background is a whiteboard with handwritten text and diagrams. On the left, it says "Ionic: Metal + Nonmetal" and shows a 3x3 grid of alternating red and blue dots. Below this, it says "Salt - NaCl" and shows a red dot labeled Na^+ and a blue dot labeled Cl^- . On the right, it says "Covalent: All Nonmetals" and shows a ball-and-stick model of a molecule. Below this, it says "Sugar - $\text{C}_6\text{H}_{12}\text{O}_6$ " and shows a ball-and-stick model of a sugar molecule. At the bottom, there are two empty rectangular boxes with wavy lines at the top, representing beakers. To the right of the video, there is a black box with white text that says "For more videos, check out: www.videochemistrytextbook.com".

Ionic: Metal + Nonmetal Covalent: All Nonmetals

Salt - NaCl

Na^+

Cl^-

Sugar - $\text{C}_6\text{H}_{12}\text{O}_6$

For more videos, check out:

www.videochemistrytextbook.com

Part 3: Apply

Apply- As a group at the Boards

Create a mindmap with labelled interconnections using the terms and concepts you have learned. Make sure it shows a good understanding of how the different intermolecular forces affect solubility.

Ensure you take a picture and insert on the next slide

Discuss your map with your teacher!



Apply: Your map

Apply - Complete this chart

Substance A	Substance B	Miscible or Immiscible?	Identify the solvent	State of final solution (s, l, g)
1L H ₂ O _(l)	1 g KCl _(s)			
10 ml H ₂ O _(l)	100 ml alcohol CH ₃ OH _(l)			
1 ml C ₆ H ₁₄ _(l)	1000 ml H ₂ O _(l)			
5 gm gelatin (think jello!)	500 ml H ₂ O _(l)			
1 ml H ₂ O _(g)	1 x 10 ⁶ L N ₂ _(g)			
100 g Cu _(s)	10 g Zn _(s)			

**Next slide(s) provide answers to
activities**

Apply - Complete this chart

Substance A	Substance B	Miscible or Immiscible?	Identify the solvent	State of final solution (s, l, g)
1L H ₂ O _(l)	1 g KCl _(s)	Miscible	H ₂ O _(l)	Liquid
10 ml H ₂ O _(l)	100 ml alcohol CH ₃ OH _(l)	Miscible	CH ₃ OH _(l)	Liquid
1 ml C ₆ H ₁₄ (l)	1000 ml H ₂ O _(l)	Immiscible	None	N/A
5 gm gelatin (think jello!)	500 ml H ₂ O _(l)	Miscible	H ₂ O _(l)	Solid
1 ml H ₂ O _(g)	1 x 10 ⁶ L N ₂ (g)	Miscible	N ₂ (g)	Gas
100 g Cu _(s)	10 g Zn _(s)	Miscible	Cu _(s)	Solid

Part 5: Beyond the course extensions {optional - none avail currently}

The Future of 3D printing with Solutions (10 mins)

